Overview of Statistical Learning and Modeling – 36-600

Lab 12T: Basic String Manipulation
Due Wednesday, November 19, 11:59pm

Trishla Nair

Data

We need some text data. Below we'll read in an inauguration speech (Joe Biden's from 2021) as formatted by the White House on its website (with one exception: I concatenated the lines containing a poem together into one line).

```
lines <- readLines("https://raw.githubusercontent.com/FSKoerner/F25-36600-data/refs/heads/main/inaugura
```

Question 1

How many lines are there? How many characters are there, overall? (This includes spaces and punctuation, for now.)

```
print(length(lines))
```

```
## [1] 211
```

```
print(nchar(lines))
```

```
##   [1] 164  22  24  26  23 100  91  81  49  21  54 248 140  64  41  75 132 105
##  [19]  89  49  48 115 109  15  16  13  14  17 102  54  76  32  43 120 106 117
##  [37] 114  55   6   6 111 138  23  53  26  19  23  46  41  26  33  35  52  20
##  [55]  39  42  34  82  30  73  66  60  33 196  24  25 147  80  22  58  58  49
##  [73]  65  63  37  33  83  62  49  72  66  10  29  17  16  28  69  60  94  55
##  [91]  35  43  17 128  32  78 144 128  34 145 191  20  21  10  13   9  88 120
## [109]  36 147  60  66  75 139  64  15  12   9   8   8   8   6  20  56  34  35
## [127] 220  80 101   9 200 105  68  43  72  89  40  54  40  98  70  68  80  76
## [145]  99  34  28 112  66  63  84  74  44 160 132  82 107   5  26  44  15  17
## [163]  29  20  28  66 102  20  10  54  21  81  29  44  84  40  69  75  71 219
## [181]  73 122  20  26  69  45  29  31  28  22  79  49  65  23  23  41  23  29
## [199]  37  27  75  18  81  87  76  62  19  21  76  53  19
```

Question 2

How many spaces are there in the speech, as formatted? (Don't worry about the fact that there would be spaces between the lines if they were all concatenated together.) One way to answer this question is to use `gregexpr()` to identify every place where there is a space, then use a for-loop to loop over the output from that function, which has elements corresponding to each line (therefore looping over lines) and count the number of spaces. For instance:

```
out   <- <the output of some function call>
total <- 0
for (ii in 1:length(lines)) {
  total <- total + length(out[[ii]])
}
```

```
num_spaces <- nchar(gsub("[^ ]", "", lines))
print(sum(num_spaces))
```

```
## [1] 2176
```

Question 3

Create a table showing how many words are on each line of the speech using `strsplit()` (for our purposes, words are separated by spaces). The output will be a list (as that of `gregexpr()`), where each element shows the individual words from a speech line. Determine the total number of words for each line, put the results in a vector, and run `table()` with that vector as input. You should find that nine of the lines have one word, four lines have two words, etc. (Note that you'll again use a for-loop, in a manner similar to the last question.)

```
split_lines <- strsplit(lines, split = " ")
word_counts_per_line <- c()
for (ii in 1:length(lines)) {
  words <- split_lines[[ii]][split_lines[[ii]] != ""]
  word_counts_per_line <- c(word_counts_per_line, length(words))
}

word_count_table <- table(word_counts_per_line)
word_count_table
```

```
## word_counts_per_line
##  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22 24 25 26 27
##  9  4 20 19 13 13  6  9 12  7 11  6 12  8  9  5 10  6  2  2  9  1  4  3  1  1
## 29 30 33 34 37 38 40 43 48
##  1  1  1  1  1  1  1  1  1
```

Question 4

Define a vector called `america` which has an element for each line that is true if the word "America" is observed in said speech line, and false otherwise. Run `sum()` on that variable to see how many lines have "America" in it. Don't overthink this: you can create this variable in one line using `grepl()`.

```
america <- grepl("America", lines)
lines_with_america <- sum(america)
print(paste(lines_with_america))
```

```
## [1] "36"
```

Question 5

Concatenate the speech into one single line. Call the output `speech`. Make sure that you insert a space between the end of each of the old lines and the beginning of the next lines. (See the usage of the `collapse` argument in `paste()`.)

```
speech <- paste(lines, collapse = " ")
substr(speech, 1, 100)
```

```
## [1] "Chief Justice Roberts, Vice President Harris, Speaker Pelosi, Leader Schumer, Leader McConnell,
```

Question 6

Working either with `lines` or with `speech`, use the framework on the last slide of the notes to remove punctuation and stopwords, leaving a single line speech in the end.

```
library(stopwords)
```

```
## Warning: package 'stopwords' was built under R version 4.4.3
```

```
speech_split <- unlist(strsplit(tolower(speech), split = "[[:space:]]+|[[:punct:]]+"))
speech_cleaned <- speech_split[speech_split != ""]
stopword.logical <- speech_cleaned %in% stopwords("en")
speech_no_stopwords <- speech_cleaned[stopword.logical == FALSE]
final_speech_line <- paste(speech_no_stopwords, collapse = " ")
substr(final_speech_line, 1, 100)
```

```
## [1] "chief justice roberts vice president harris speaker pelosi leader schumer leader mcconnell vice
```

Question 7

What are the top 20 words (meaning, non-stopwords) in Biden's speech? You might think that "America" appears less than you'd expect, given your result from Question 4. But, when you searched for "America" above, you'll note that "American", "Americans", and variety of other formulations with the same root appeared, and all counted. (Unless you crafted a really exact regex!)

```
# Rerunning the code from Question 6 to ensure the dependencies are available
word_frequencies <- table(speech_no_stopwords)
sorted_frequencies <- sort(word_frequencies, decreasing = TRUE)
print(sorted_frequencies[1:20])
```

```
## speech_no_stopwords
##        us   america       can         s       one    nation democracy      must
##        27        20        18        17        15        14        11        10
##  american americans   another    people     story     today      know      much
##         9         9         9         9         9         9         8         8
##     unity     world   history president
##         8         8         7         7
```